

Subject Description Form

Subject Code	ITC426ME
Subject Title	Business Intelligence and Analytics for Fashion
Credit Value	3
Level	4
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	In a data-driven business environment, the ability to convey hard messages with clever visualization is essential and valuable. This subject aims to • Introduce the major concepts of business intelligence and data mining; • Provide knowledge about how to provide business solution using business intelligence techniques; • Explain the roles that software tools and applications play in business intelligence and data mining; • Give students an overall understanding the best practices for building business intelligence projects successfully for fashion related application.
Intended Learning Outcomes <i>(Note 1)</i>	Upon completion of the subject, students will be able to: a. Recognize the need for business intelligence to support management of fashion business; b. Apply the concepts and techniques of business intelligence and analytics in business practice; c. Apply information technologies for data mining; d. Analyze the requirements for management support and identify appropriate tools and techniques for data mining in order to create innovative business intelligence solutions; e. Acquire and innovatively apply skills in using Microsoft Excel and other menu based visualization tools to create compelling visual report and analysis; and f. Acquire and innovatively apply skills in using programmable visualization tools (R packages) to create flexible visual presentations.
Subject Synopsis/ Indicative Syllabus <i>(Note 2)</i>	(I) Introduction to business intelligence (BI) An overview of the subject topic of business intelligence; compare different jargon of datamining, machine learning, business intelligence, multiple dimensional data analysis. (II) Big Data Understand the principles, methods and technologies of big data management for business improvements and innovations. (III) Data warehousing and Data visualization Introduction to multidimensional data management. Students will learn concepts, methods, and applications of data visualization methods. The

	course will introduce interesting examples in different application areas. Students will also learn a spectrum of visualization tools from simple GUI-based packages such as Tableau, Excel, Pajek to more advanced programmable visualization packages in R language. They will be guided in creating engaging and interactive visualizations. (IV) Data mining; Machine learning; Core ideas in data mining; the steps in data mining; using R for data mining Hands-on activities using database management systems, SQL and business intelligence software to perform data mining techniques of classification, clustering, and association rule mining.																																																						
Teaching/Learning Methodology (Note 3)	Lecture discussion and tutorials will underpin students’ understanding of the subject contents. Instrumental cases will be adopted to illustrate the usability of the principles in today’s global fashion business. Apart from lecturing, coursework including assignments, online quizzes and group project will also be key parts of the knowledge learning.																																																						
Assessment Methods in Alignment with Intended Learning Outcomes (Note 4)	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="6">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th><th>e</th><th>f</th></tr><tr><td>1. Assignments and participations</td><td>20%</td><td>✓</td><td></td><td>✓</td><td></td><td>✓</td><td>✓</td></tr><tr><td>2. Quizzes</td><td>15%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>3. Projects</td><td>25%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>4. Examination</td><td>40%</td><td>✓</td><td>✓</td><td>✓</td><td>✓</td><td></td><td></td></tr><tr><td>Total</td><td>100 %</td><td colspan="6"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: 20% is given for student’s assignments and participation in terms of quality of questions, answers and student engagement in both lectures and tutorials Online quizzes will be provided to assess students’ understanding on subject matters. All quizzes are open book. There are either multiple choice questions and/or short questions. Students are advised NOT to click on quizzes link until you are ready to take the quiz. There is no retakes, you can only take the quiz once. The subject has a group project, which requires the students to develop a social media enabled strategy and marketing plan to promote a fashion brand. The project requires a project proposal and presentation as well. A final 2-hour exam will be closed book examination. “The materials submitted for this assessment must be the student’s own work. The submitted work may not be accepted for the purpose of assessment if its authenticity is questionable. Submitting GenAI-generated materials as students’ own work or part of their work is an act of academic dishonesty. Students who are found committing academic dishonesty will face disciplinary actions.”	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed						a	b	c	d	e	f	1. Assignments and participations	20%	✓		✓		✓	✓	2. Quizzes	15%	✓	✓	✓	✓			3. Projects	25%	✓	✓	✓	✓	✓	✓	4. Examination	40%	✓	✓	✓	✓			Total	100 %						
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Student Study Effort Expected	Class contact:	
	▪ Lecture	26 Hrs.
	▪ Tutorial	12 Hrs.
	Other student study effort:	
	▪ Self-study/Preparation	46 Hrs.
	▪ Project/Assignments	24 Hrs.
	Total student study effort	108 Hrs.
Reading List and References	<u>Books</u> Fischetti, T. & Lantz, B. (2016). R: Data Analysis and Visualization, Packt Publishing. Yau, N. (2011). Visualize this: the Flowing Data guide to Design, Visualization, and Statistics. John Wiley & Sons. Few, S. (2012). Show Me the Numbers: Designing Tables and Graphs to Enlighten, Analytics Press. Linoff, G. S., & Berry, M. J. (2011). Data Mining Techniques: for Marketing, Sales, and Customer Relationship Management, 3rd edition. John Wiley & Sons. Provost, F., & Fawcett, T. (2013). Data Science for Business: What You Need to Know About Data Mining and Data-Analytic Thinking. O'Reilly Media, Inc. Shmueli, G., Patel, N. R., & Bruce, P. C. (2011). Data Mining for Business Intelligence: Concepts, Techniques, and Applications in Microsoft Office Excel With XLminer. John Wiley and Sons.	

